



Bahria University, Islamabad

Department of Software Engineering

Computer Communication & Networking Lab
(Fall-2025)

Instructor: Engr. Muhammad Faisal Zia

Student: Malik Shajar Hussain

Enrollment: 01-131232-043

Lab Journal: 07

Date: 14/10/2025

Task No:	Task Wise Marks		Documentation Marks		Total Marks (20)
	Assigned	Obtained	Assigned	Obtained	
1					
2					
3					
4					
5					

Comments:

Signature

LAB # 7

Building and testing a Router based network

Objectives

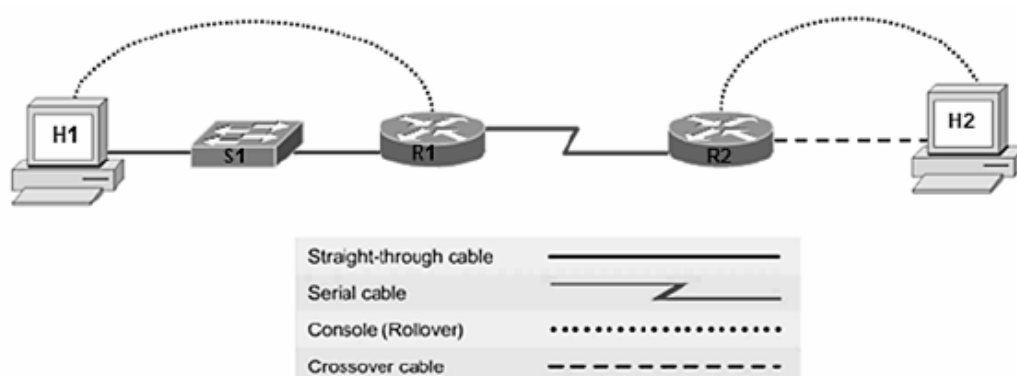
- Configure the device host name for a router.
- Configure Ethernet and serial interfaces, including description.
- Configure a message of the day (MOTD) banner.
- Verify connectivity between hosts and routers.

Introduction

This is a lab to review previously covered IOS commands. In this lab, you will cable the equipment as shown in the topology diagram. You will then configure the devices to match the addressing table. After the configurations have been saved, you will verify your configurations by testing for network connectivity.

After the devices have been configured and network connectivity has been verified, you will use IOS commands to retrieve information from the devices to answer questions about your network equipment.

This lab provides minimal assistance with the actual commands necessary to configure the router.



Device	Host Name	Interface	IP Address	Subnet Mask
R1	R1	Serial 0/0/0 (DCE)	172.17.0.1	255.255.0.0
		FastEthernet 0/0	172.16.0.1	255.255.0.0
R2	R2	Serial 0/0/0 (DTE)	172.17.0.2	255.255.0.0
		FastEthernet 0/0	172.18.0.1	255.255.0.0

Required Resources

The following resources are required:

- Two routers, each with an Ethernet and serial interface.

- Two Windows XP computers with HyperTerminal installed
- Two straight-through Category 5 Ethernet cables (H1 to S1 and S1 to R1)
- Crossover Category 5 Ethernet cable (H2 to R2)
- Null serial cable (R1 to R2)
- Console cables (H1 to R1 and H2 to R2)
- Access to the host H1 and H2 command prompt
- Access to the host H1 and H2 network TCP/IP configuration

From each host computer, start a HyperTerminal session to the attached router.

Connection	Cable Type	Description
H1 → S1	Straight-through Ethernet cable	Connects Host H1 to Switch S1
S1 → R1 (FastEthernet 0/0)	Straight-through Ethernet cable	Connects Switch S1 to Router R1
R1 (Serial 0/0/0 DCE) → R2 (Serial 0/0/0 DTE)	Serial (null serial) cable	Connects R1 and R2 via WAN serial link
H2 → R2 (FastEthernet 0/0)	Straight-through Ethernet cable	Connect Host H2 directly to Router R2
H1 → R1 (Console port)	Console (rollover) cable	Provides console access from H1 to R1
H2 → R2 (Console port)	Console (rollover) cable	Provides console access from H2 to R2

Step 1: Configure host computer IP settings.

- Make sure that the host computers are connected according to the topology diagram.
- Configure the hosts with static IP addresses using the following settings.

H1 attached to the S1 switch:

IP address: 172.16.0.2
 Subnet mask: 255.255.0.0
 Default gateway: 172.16.0.1

H2 attached to R2 directly:

IP address: 172.18.0.2
 Subnet mask: 255.255.0.0
 Default gateway: 172.18.0.1

Step 2: Log in to each router and configure the basic settings.

Note: Perform each step for both routers.

- a. Configure a host name for each of the two routers.

```
Router>enable
Router#configure terminal
Router(config)#hostname R1
```

Note: Use **R2** for the name of the second router.

- b. Configure a message-of-the-day (MOTD) banner using the **banner motd** command. When a user connects to the router, the MOTD banner appears before the login prompt. In this example, the number sign (#) is used to start and end the message. The # is converted to ^C when the running-config is displayed.

```
R1(config)#banner motd #Unauthorized Use Prohibited#
```

Step 3: Configure the serial interface on R1.

In global configuration mode, configure serial interface 0/0/0 on R1. See the Router Interface Summary table at the end of the lab for the proper designation of the serial interface on the router that you are using. Because the R1 serial 0/0/0 interface is acting as the DCE for the WAN link, it is necessary to configure the clock rate. When configuring an interface, always use the **no shutdown** command to enable it.

```
R1(config)#interface serial 0/0/0
R1(config-if)#description WAN link to R2
R1(config-if)#ip address 172.17.0.1 255.255.0.0
R1(config-if)#clock rate 64000
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config-if)#exit
```

Step 4: Configure the serial interface on R2.

In global configuration mode, configure serial 0/0/0 on router R2. See the Router Interface Summary table at the end of the lab for the proper designation of the serial interface on the router that you are using.

```
R2(config)#interface serial 0/0/0
R2(config-if)#description WAN link to R1
R2(config-if)#ip address 172.17.0.2 255.255.0.0
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#exit
```

Step 5: Verify that the serial connection is functioning.

- a. Use the **ping** command to test connectivity to the serial interface of the other router. From R1, ping the R2 router serial interface.

```
R1#ping 172.17.0.2
```

Does the ping work? __Yes____

b. From R2, ping the R1 router serial interface.

```
R2#ping 172.17.0.1
```

Step 6: Configure the Fast Ethernet interface on R1.

In global configuration mode, configure the Fast Ethernet 0/0 interface on router R1. See the Router Interface Summary table at the end of the lab for the proper designation of the Ethernet interface on the router that you are using.

```
R1(config)#interface FastEthernet 0/0
R1(config-if)#description R1 LAN Default Gateway
R1(config-if)#ip address 172.16.0.1 255.255.0.0
R1(config-if)#no shutdown
```

Step 7: Configure the Fast Ethernet interface on R2.

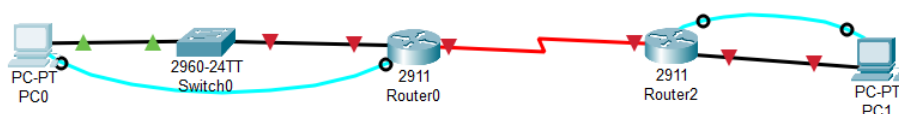
In global configuration mode, configure the Fast Ethernet 0/0 interface on R2. Refer to the Router Interface Summary table at the end of the lab for the proper designation of the Ethernet interface on the router that you are using.

```
R2(config)#interface FastEthernet 0/0
R2(config-if)#description R2 LAN Default Gateway
R2(config-if)#ip address 172.18.0.1 255.255.0.0
R2(config-if)#no shutdown
R2(config-if)#exit
```

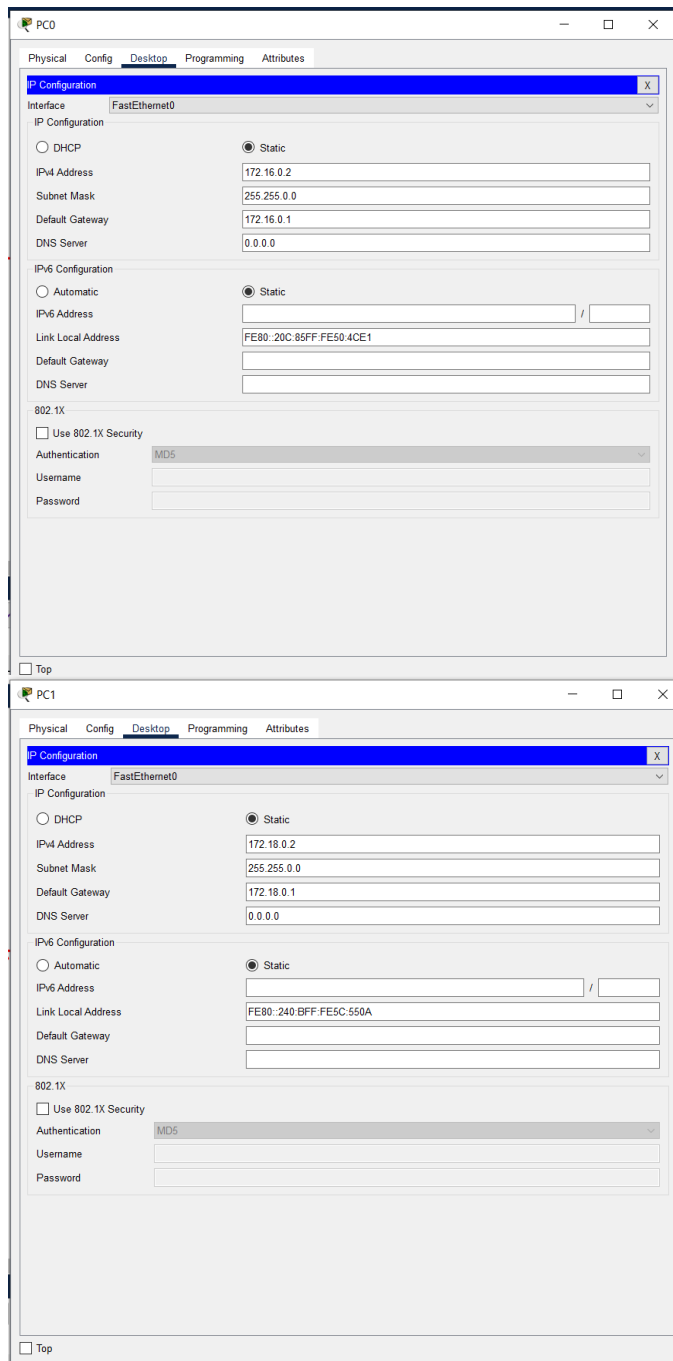
Tasks

Task 1

Perform the set of tasks performed above.



1) Configure the two PCs (hosts)



2) Basic setup on both routers (R1 and R2)

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R
Router(config)#hostname R1
R1(config)#banner motd #Unauthorized Use Prohibited#
R1(config)#

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R2
R2(config)#banner #Unauthorized Use Prohibited#
^
% Invalid input detected at '^' marker.

R2(config)#banner motd #Unauthorized Use Prohibited#
R2(config)#
```

3) Configure the serial link R1↔R2

```
Unauthorized Use Prohibited

R1>enable
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface serial 0/0/0
R1(config-if)#description WAN link to R2
R1(config-if)#ip address 172.17.0.1 255.255.0.0
R1(config-if)#clock rate 64000
R1(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to down
R1(config-if)#exit
R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#disable
R1>exit

Unauthorized Use Prohibited

R2>enable
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#interface serial 0/0/0
R2(config-if)#description WAN link to R1
R2(config-if)#ip address 172.17.0.2 255.255.0.0
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
```

4) Verify the serial link

```
Unauthorized Use Prohibited

R1>enable
R1#ping 172.17.0.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.17.0.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/12/40 ms

R1#
```

```
R2>enable
R2#ping 172.17.0.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.17.0.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/8/39 ms

R2#
```

5) Configure the routers' LAN (Ethernet/Gigabit) interfaces

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface GigabitEthernet 0/0
R1(config-if)#description R1 LAN Default Gateway
R1(config-if)#ip address 172.16.0.1 255.255.0.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
```

```

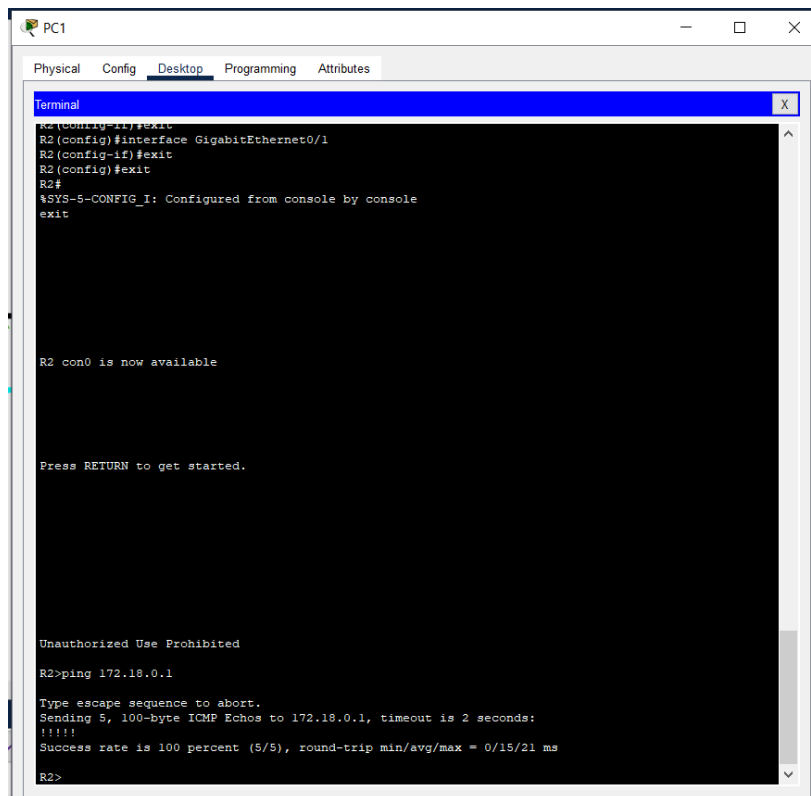
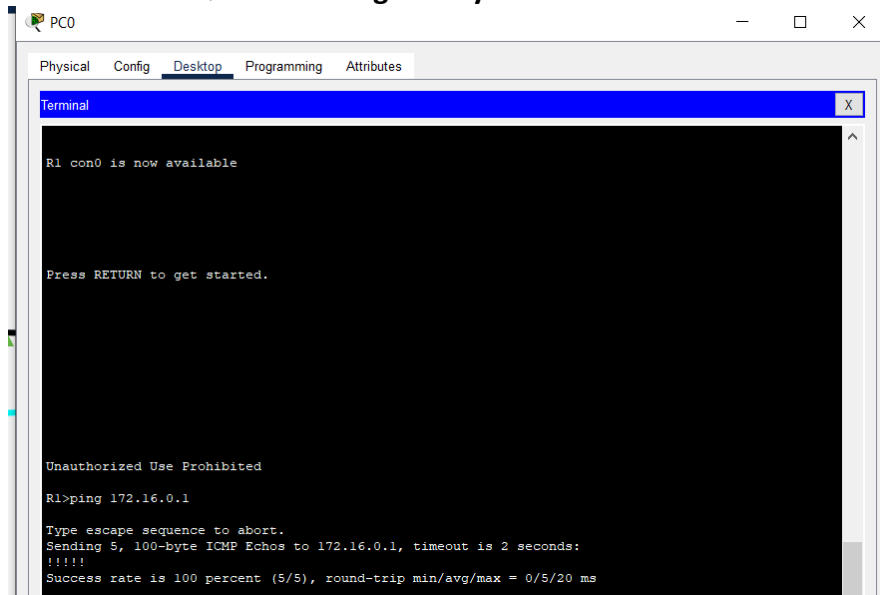
R2(config)#interface GigabitEthernet 0/0
R2(config-if)#description R1 LAN Defeault Gateway
R2(config-if)#ip address 172.18.0.1 255.255.0.0
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

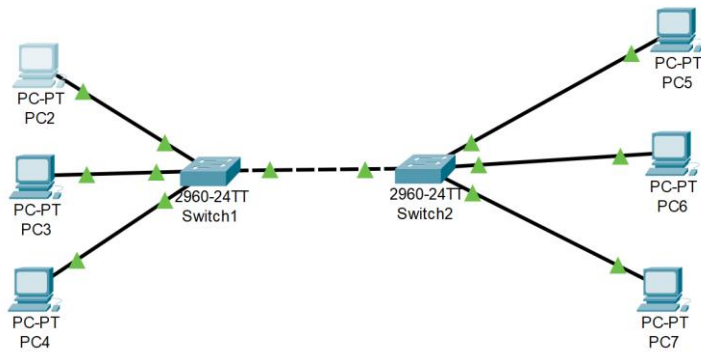
```

From each PC → its default gateway



Task 2

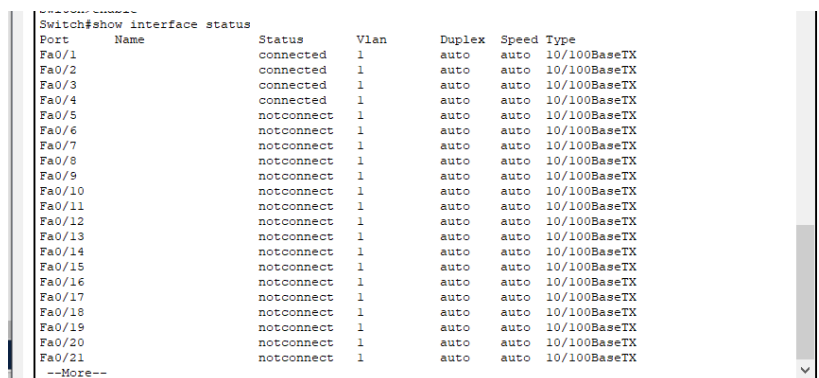
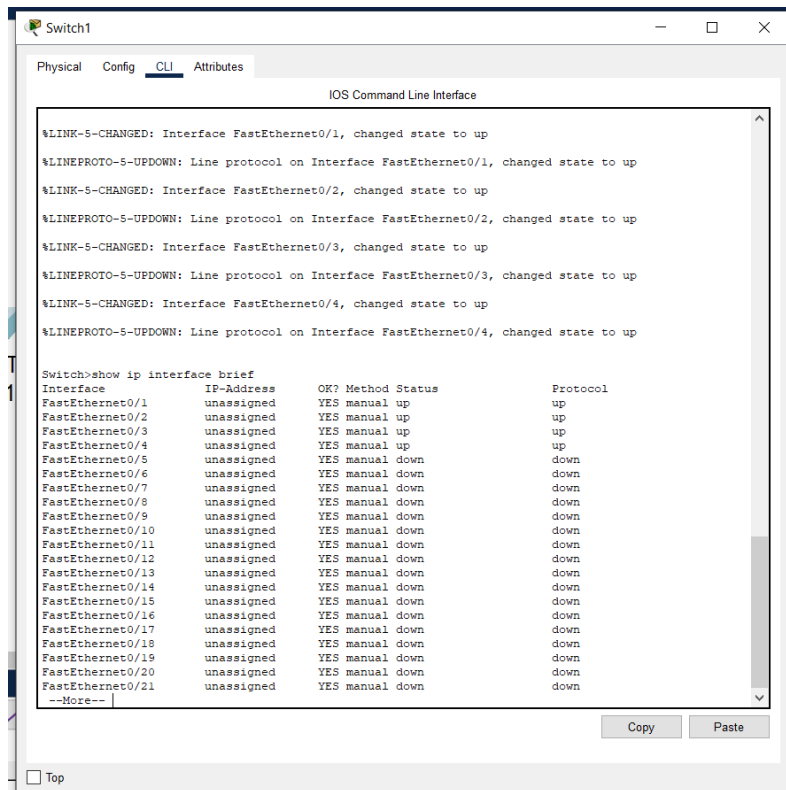
Create a network by connecting 2 switches together with three Host PCs connected to each switch.



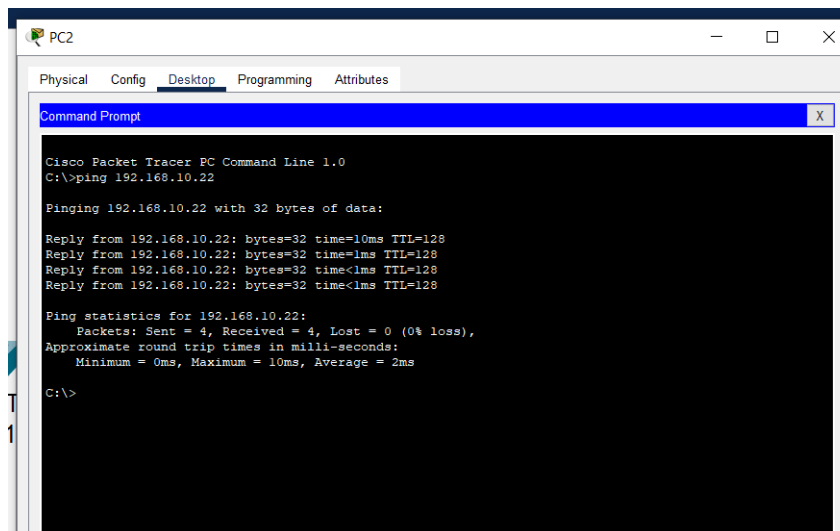
Setting the IP:

- **PC1:** 192.168.10.11
- **PC2:** 192.168.10.12
- **PC3:** 192.168.10.13
- **PC4:** 192.168.10.21
- **PC5:** 192.168.10.22
- **PC6:** 192.168.10.23

BRIEF STATUS OF SWITCH PORTS:



PING CONNECTION AMONG PC,S:



SIMULATION OF PACKETS AMONG TWO PC'S:

PDU List Window												
Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete		
	Successful	PC2	PC6	ICMP		0.000	N	0	(edit)	(delete)		

Observation (Task 1 & Task 2)

1. Two routers were configured with serial and LAN interfaces, enabling point-to-point and local connectivity.
2. Proper IP addressing and clock rate settings established communication across the router-to-router link.
3. PCs were assigned IPs and gateways to verify end-to-end connectivity through the routers.
4. Two switches were interconnected, and three host PCs were attached to each using straight-through cables.
5. All PCs were configured in the same subnet, confirming Layer-2 communication through the switches.
6. Successful ping tests between all devices verified correct cabling, addressing, and functional data flow in both LAN and WAN segments.